Entering the Movie Business: A Data-Driven Approach

Insights for new movie studio
By Group 8

Group Members

Our team was composed of 4 data analysts who worked on the data. They include;

- Vihaan Sheth
- Kevin Kibet
- Vincent Ngochoch
- Chris Gitonga

Business understanding

Problem Statement

- As the company now sees all the big companies creating original video content and they want to get in on the fun. They have decided to create a new movie studio, but since they don't know anything about creating movies. We were charged with exploring what types of films are currently doing the best at the box office. So as to maximize profits and reduce the risks associated with entering this sector.

- By analyzing different movie datasets we will identify. Which movie genres tend to receive the highest ratings?, Which genres have the highest return on investment (ROI), Which film languages are associated with higher popularity? and Are certain directors more likely to deliver high foreign box office success?
- This will ensure a strategic and well informed entry into the movie sector

Data understanding

What Datasets are we using and why?

- To make informed business decisions we needed reliable data sources hence we obtained our data from the following databases;1) The Box Office Mojo database, 2) IMDB(SQLite database), 3) Rotten Tomato Movies database, 4) The movies Database, 5) The Numbers Database.
- This datasets will allow us to identify what types of films to prioritize and produce for maximum profits.

How Will The Data Help Us Minimize Risks?

- By analyzing the data from the different databases it guides in making smart investments in film production. Factors such as identifying high-performing genres, directors, and languages with proven track records of success can reduce the company's financial uncertainty by prioritizing films with highest commercial potential.

How Does This Data Support Our Business Strategy?

- Entering the film industry requires data-driven creative decisions. By analyzing historical box office performance, we ensure our production choices are evidence-based, helping us select films that align with both audience demand and commercial viability.

Business Objectives

Audience and critic rating trends by Genre

- Identifying genres with consistently high critic and audience ratings helps reduce creative risk and enhances confidence in commercial success.
- Investing in genres with a track record of high ratings positions the studio for sustainable growth while prioritizing artistic credibility and audience satisfaction

ROI trends in terms of Genre

- Identifying genres with consistently high ROI helps minimize financial risk and maximizes profitability for new market entrants.

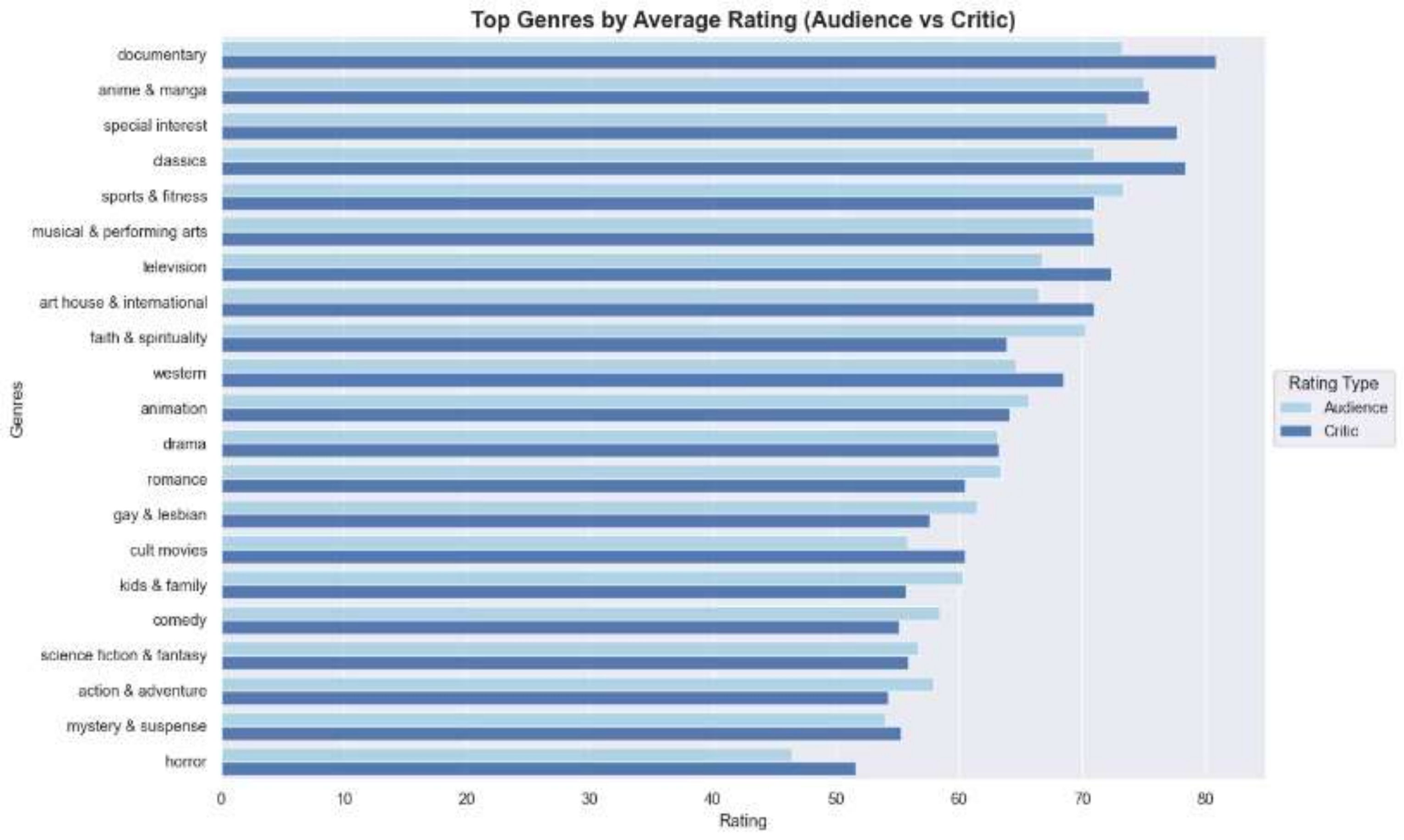
Relationship between Language and Popularity.

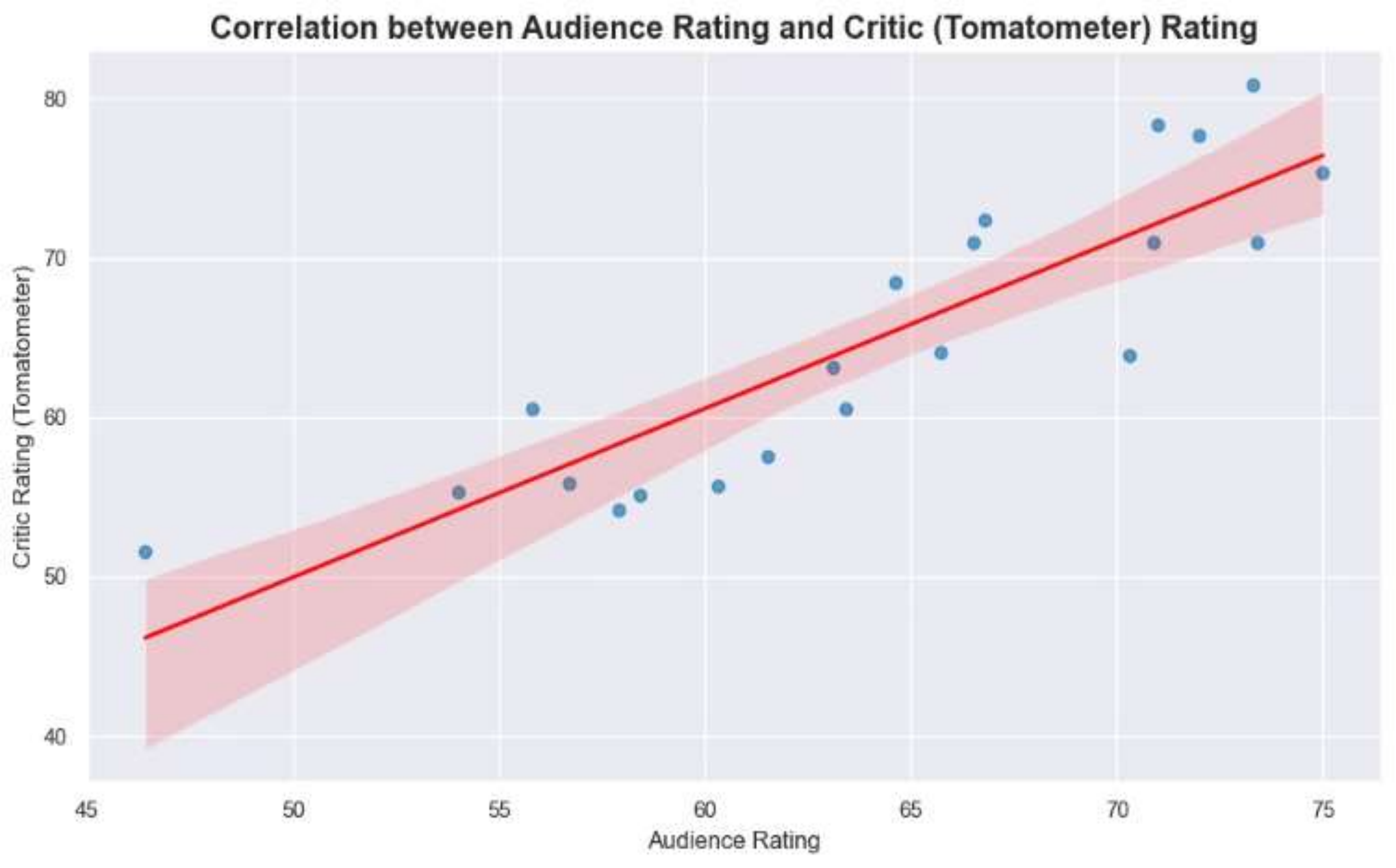
- Identifying which languages dominate the industry movie aids strategic decision-making by revealing global market opportunities and minimizing financial risk.
- Determining the popularity distribution of the top languages used in movie production and identifying what language has highly rated popularity votes will aid in determining what language to use in movie production in order to increase our chances of getting a high popularity score

Relationship between Directors and Worldwide Gross Earnings

- Identifying which directors have the highest worldwide gross earnings helps pinpoint who consistently deliver success allowing for smarter partnership decisions.
- Tracking director performance trends over time aids in understanding how a director's' box office power has evolved helping us identify rising stars and fading trends in audience preferences

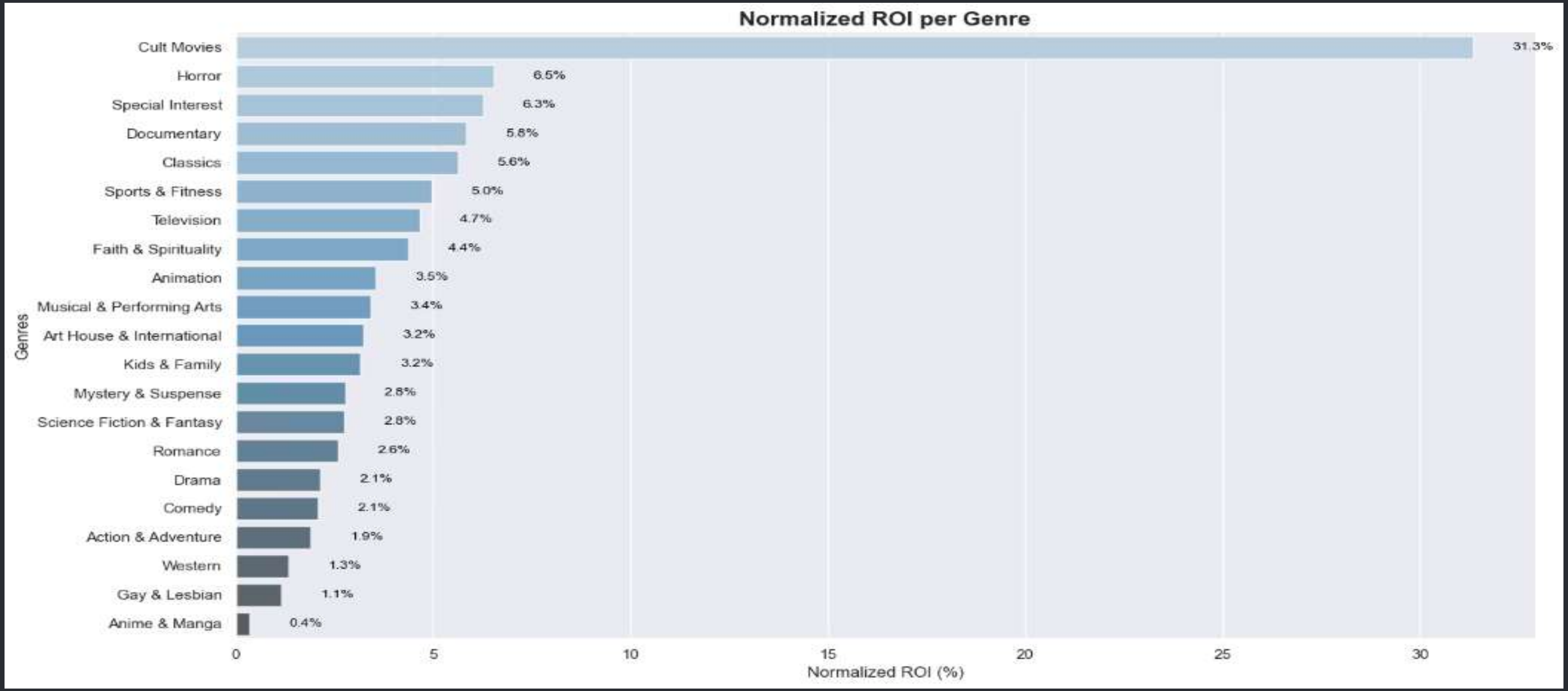
Audience and critic rating trends by Genre





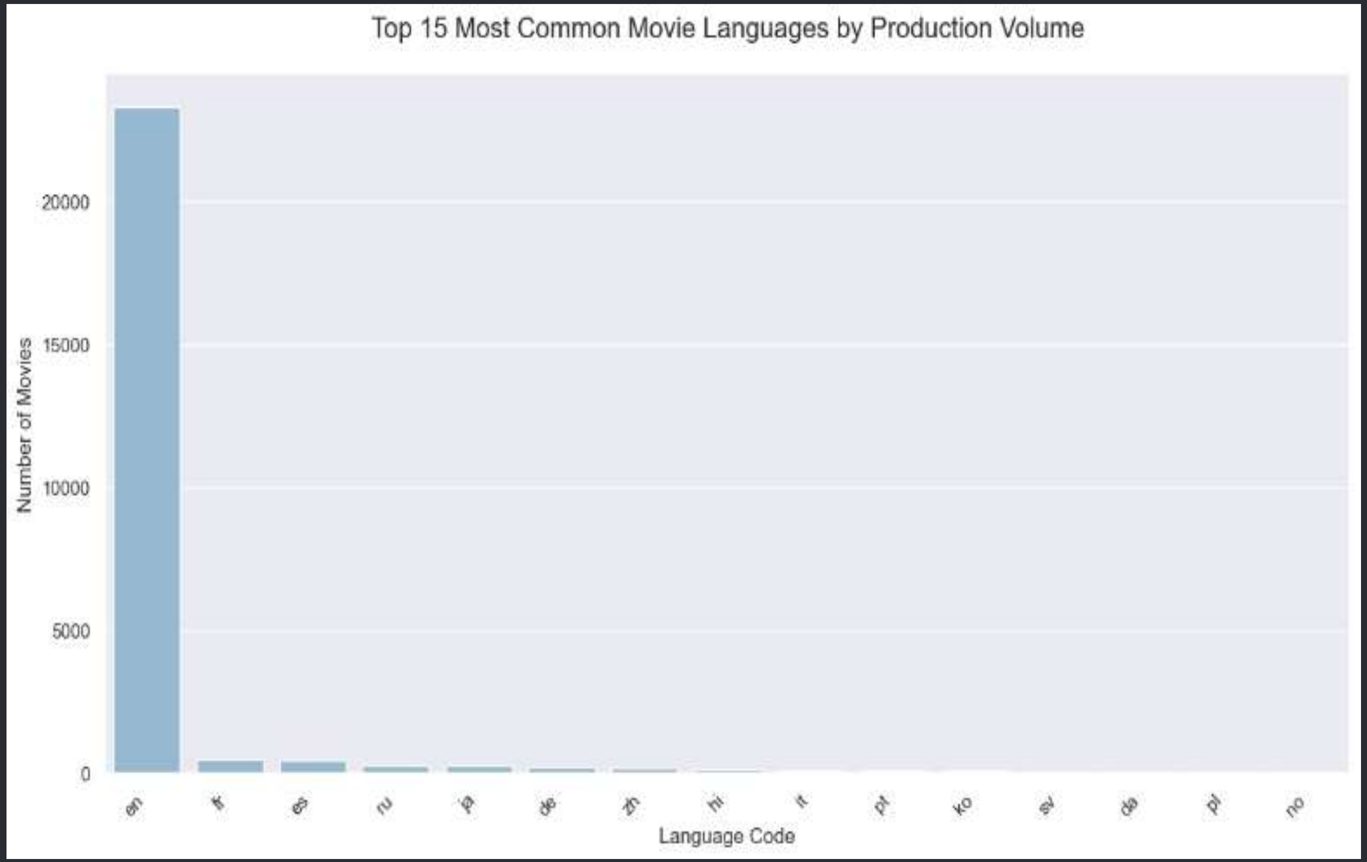
- **Documentaries** stand out as the highest-rated genre by critics, and they also perform very well with audiences.
- **Anime & Manga, Special Interest,** and **Classics** are also among the top genres appreciated by both critics and audiences, indicating strong dual-market appeal.
- **Television** receives relatively high audience ratings but lower critic scores, suggesting it may appeal more to popular tastes than to critical standards.
- Genres like **Faith & Spirituality** and **Western** tend to score lower across both metrics, potentially indicating limited broad appeal or niche value.
- **Sports & Fitness** shows an interesting case where audience ratings surpass critic scores, implying mass appeal despite critical reservations.
- From the scatter plot the correlation suggests that aligning with audience preferences does not come at the cost of critical acclaim—genres that perform well with audiences are likely to satisfy both sides.

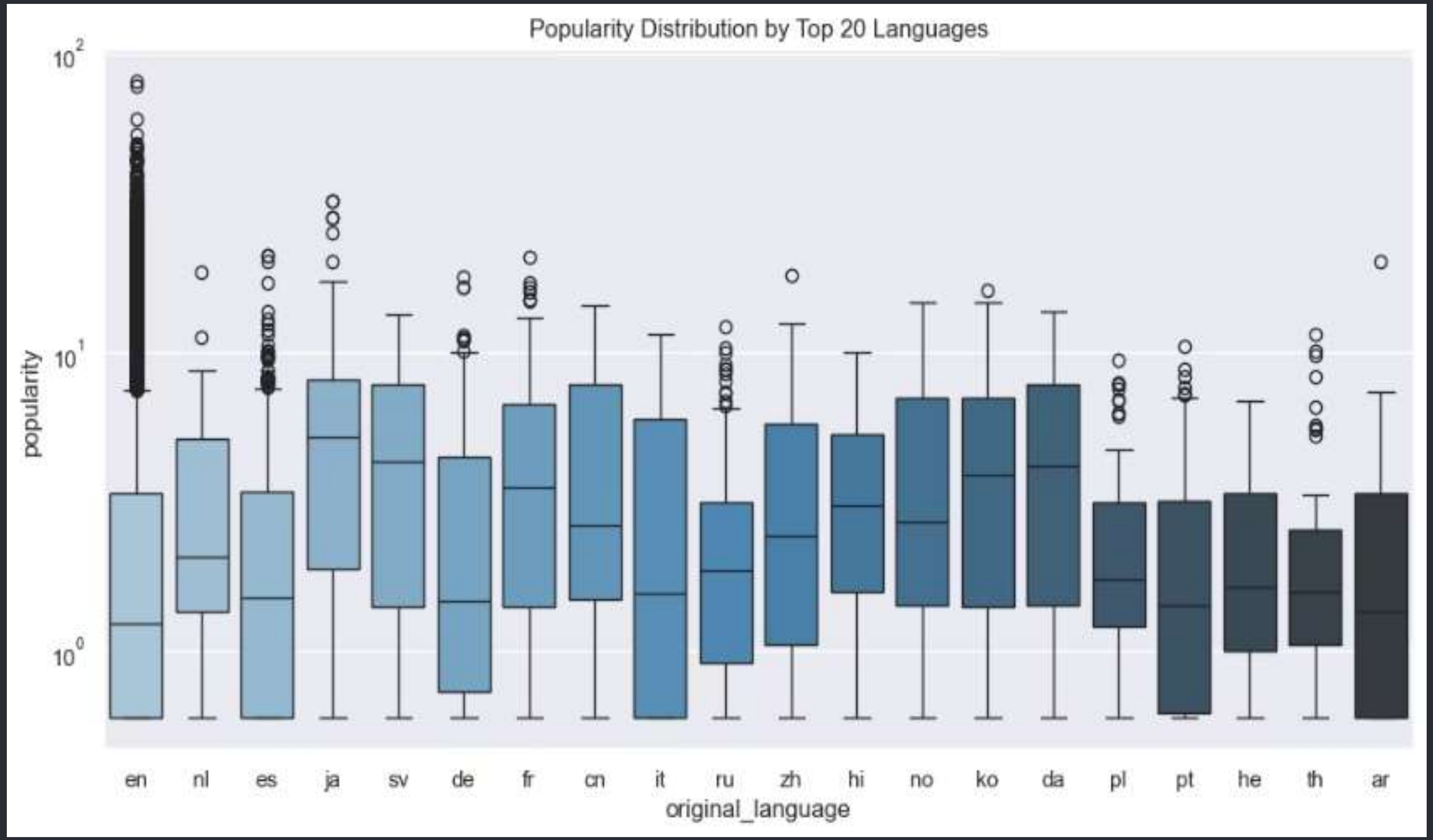
ROI trends in terms of Genre



- Identifying genres with consistently high ROI helps minimize financial risk and maximizes profitability for new market entrants.
- Cult Movies stand out with a significantly higher normalized ROI (~31.3%) compared to all other genres. This suggests that despite often lower production budgets, these films generate exceptional returns — likely due to niche, loyal audiences and strong word-of-mouth.
- Genres like Horror (6.5%), Special Interest (6.3%), and Documentary (5.8%) also perform well in terms of ROI. These typically have lower budgets and focused audiences, which makes them efficient from an investment standpoint.
- Traditional commercial genres such as Action & Adventure, Comedy, and Drama have lower normalized ROI values (around 2%), indicating that high production costs may dilute their profitability even if box office performance is strong.
- Genres with niche or limited mainstream appeal, such as Western, Gay & Lesbian, and Anime & Manga, tend to have the lowest ROI, suggesting limited reach or returns relative to investment.
- By prioritizing genres with high ROI and focused audience appeal, the studio can build a sustainable production strategy that aligns artistic goals with strong financial outcomes

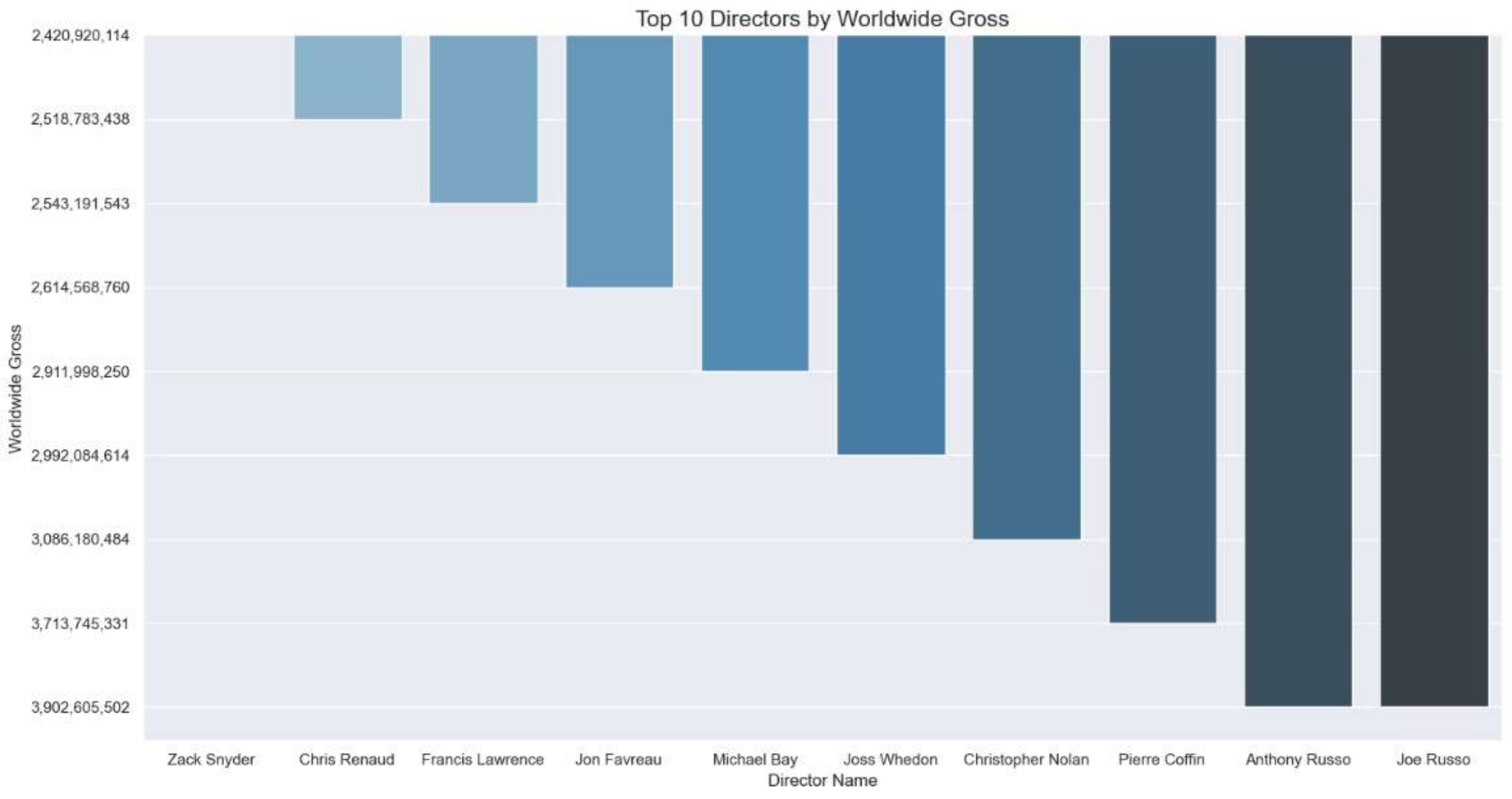
Relationship between Language and Popularity





- English (en) dominates with over 20,000 movies, far exceeding any other language, followed by French (fr) and Spanish (es).
- This trend indicates that most studios tend to produce content in English, which may be due to its global acceptance, availability of talent, and ease of distribution. For a new studio, prioritizing English-language productions may align with industry norms and maximize early traction
- In the **Popularity Distribution boxplot** we focused on the outliers because, Languages that with frequent or extreme outliers may occasionally produce global blockbusters and highlight potential for breakout success, even in languages with low production volume.
- English (en) not only leads in production but also shows a wide range of popularity with multiple strong outliers, indicating many top-performing movies.
- Japanese (ja), Spanish (es), French (fr), and Russian (ru) also exhibit strong outliers, signaling their ability to produce highly popular films despite smaller outputs.

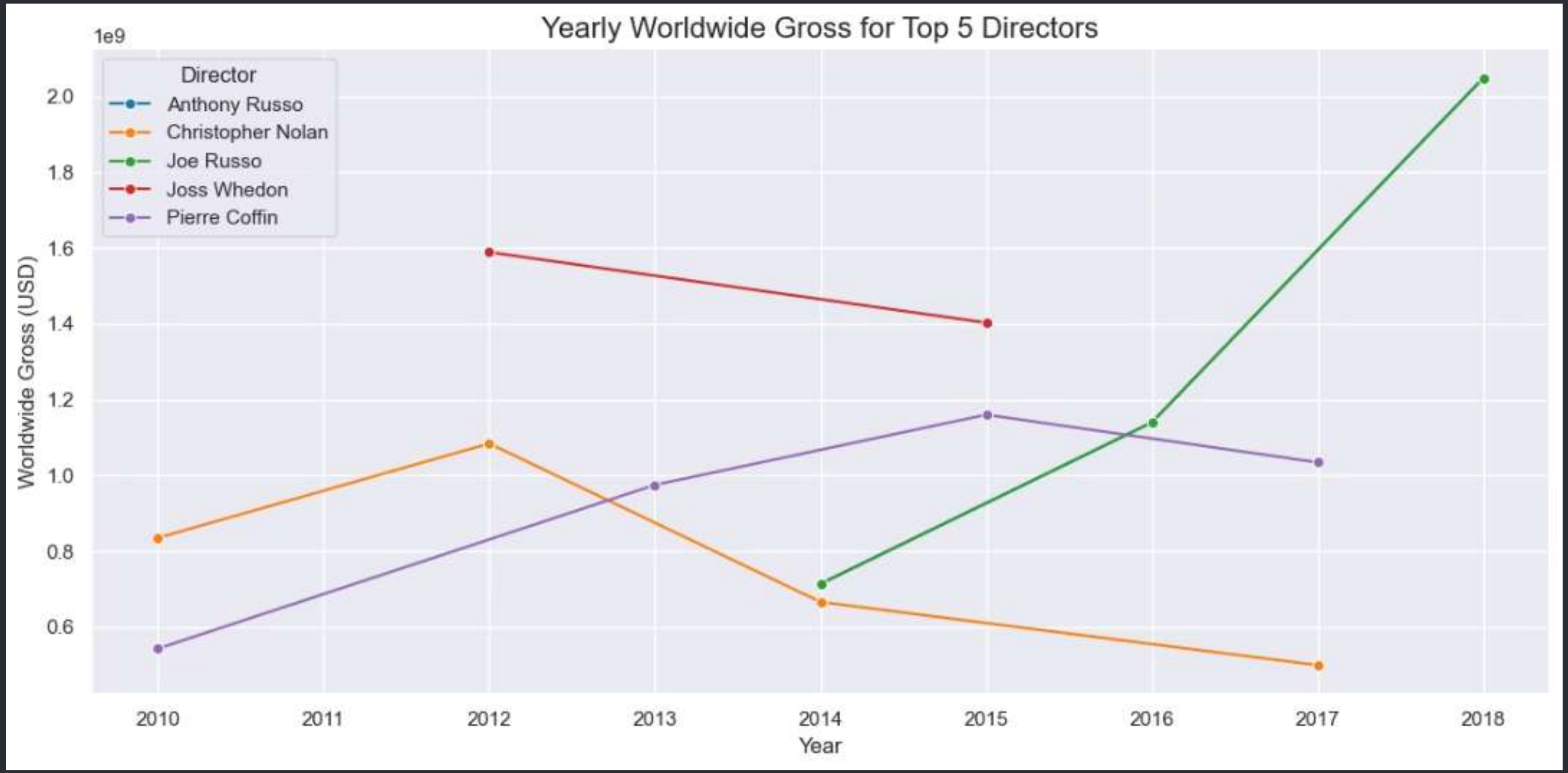
Relationship between Directors and Worldwide Gross Earnings



From the top 10 directors by worldwide gross bar graph we see that:

- Joe Russo and Anthony Russo dominate the chart, both nearing or exceeding the $3.9 billion mark. This reflects their significant contributions to blockbuster franchises, particularly within the Marvel Cinematic Universe.
- Pierre Coffin, known for animated films like Despicable Me and Minions, ranks just below, indicating the strong earning potential of animated family films.
- Christopher Nolan stands out as the only director in the top tier known for original, high-concept blockbusters, suggesting that originality paired with cinematic scale can also yield massive returns.
- Directors like Michael Bay, Joss Whedon, and Jon Favreau reflect the commercial power of action-heavy, effects-driven films and franchise-building.
- From the data, there is a clear trend in that: franchise integration, visual spectacle, and genre appeal (especially action, animation, and superhero films) consistently lead to high global box office success including amazing directors.

Relationship between Directors and Worldwide Gross Earnings



Tracking director performance trends over time aids in understanding how a director's box office power has evolved helping us identify rising stars and fading trends in audience preferences

Key insights from yearly gross trends:

1. **Franchise Peaks**

   Russo Brothers and Whedon show sharp revenue spikes during Marvel releases, proving franchise films drive predictable blockbuster earnings.

2. **Animation's Steady Growth**

   Coffin's incremental rises highlight animated sequels' reliable revenue stream compared to live-action volatility.

3. **Director Consistency**

   Nolan's stable performance demonstrates original films can yield steady returns without franchise dependence.

4. **Strategic Timing Matters**

   Franchise gaps (e.g., Russo Bros' dips between Marvel films) reveal the need for planned release cycles to sustain momentum.

Business recommendations

1 Focus on High-Impact Genres

- Prioritize genres with both critical and commercial strength (e.g Documentaries, Cult Films)
- Use low-budget, high-ROI genres (e.g., Horror) for early-phase projects.
- Avoid low-impact genres unless justified by unique branding or demand.

2 Balance Ratings with Financial Returns

- Evaluate genres through both critical and financial lenses before greenlighting.
- Combine high-ROI genres with well-reviewed ones to balance credibility and revenue
- Stay agile by updating the genre mix as market and rating trends evolve.

3 Language strategy for global reach

- Begin with English-Language Productions for maximum reach and recognition.
- Plan multilingual or subtitled releases to expand into regional markets..
- Adapt to language trends to stay competitive in evolving markets.

4 Leverage Proven Directional Talent

- Collaborate with or learn from high-grossing directors to fast-track success.
- Use director branding as a marketing asset to attract audiences and investors.
- Scout emerging talent with similar creative styles to balance cost and quality.
- Track director performance over time to identify long-term value creators.
- .

Next Steps

1. Commission pilot projects in high-ROI genres.
2. Engage consultants or advisors from high-performing production backgrounds.
3. Reassess strategy bi-annually with updated data and market trends.

THANK YOU

Any questions ?

GROUP 8:

I. Vihaan Sheth
II. Kevin Kibet
III. Vincent Ngochoch
IV. Chris Gitonga